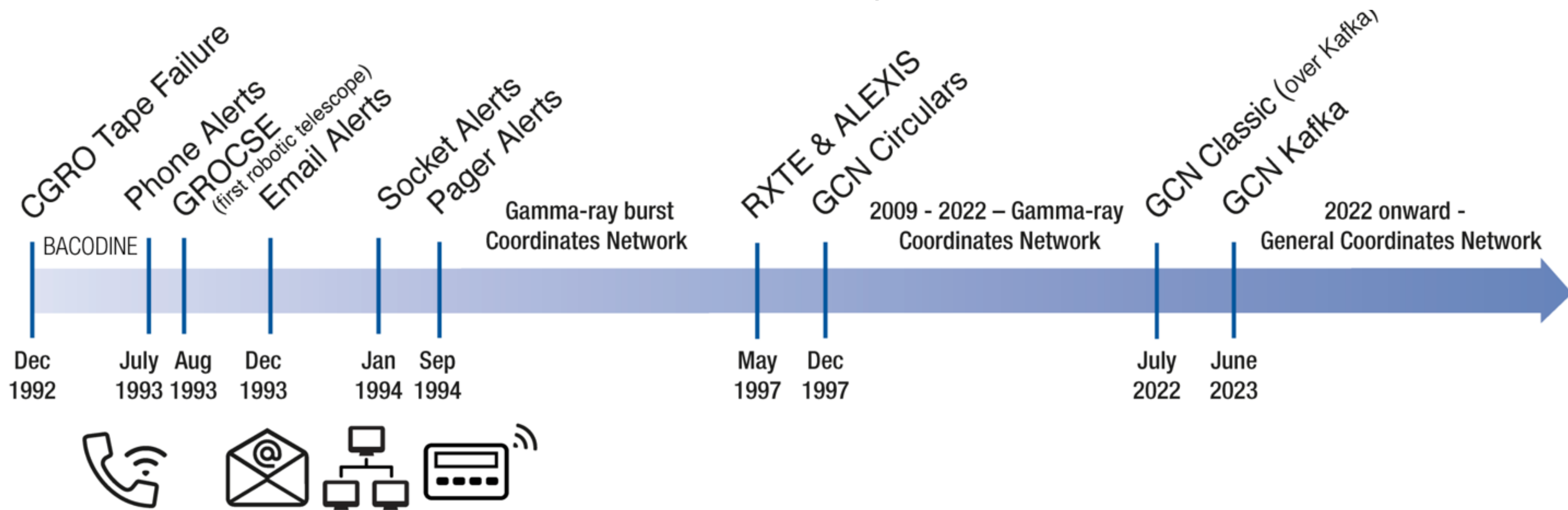


Historical Background



- **1993:** GCN established by NASA's Goddard Space Flight Center.
- Initially connected **CGRO/BATSE** with ground-based observatories. To enable immediate follow-up by ground based robotic telescopes. Technology based on phone calls, mails, alerts via internet
- **1990s–2000s:** Expansion to include **Swift**, **INTEGRAL**, **Fermi-GBM**, and other missions. Still focused on gamma-ray astronomy
- Finally extended to the full Time-Domain and multi-messenger community, including gravitational waves, neutrinos, fast radio bursts, SN, optical transients, FBOTs...

Some GCN-enabled milestones

- **1999 – GRB 990123**

First prompt optical flash detected from a GRB (ROTSE).

GCN enabled near real-time follow-up within seconds of the BATSE alert.

- **2017 – GW170817 / GRB 170817A**

First gravitational-wave and gamma-ray joint detection (LIGO/Virgo + Fermi/INTEGRAL).

GCN distributed coordinated alerts worldwide within minutes.

Marked the birth of multi-messenger astronomy.

- **2017 – IceCube-170922A**

First high-energy neutrino linked to an astrophysical source (blazar TXS 0506+056).

GCN alerts unified neutrino, X-ray, gamma-ray, and optical observations.

- **2022 – GRB 221009A (“The BOAT”)**

The Brightest Of All Time, detected across the spectrum.

GCN alerts allowed rapid follow-up from Fermi, Swift, INTEGRAL, H.E.S.S., MAGIC, and others.

Showcased the power of modern GCN for real-time global coordination.